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be reached by the actual stock. But in the simple example of a cup of coffee, the drink eventually will reach room temperature.

Runaway Loops—Reinforcing Feedback

> I'd need rest to refresh my brain, and to get rest it's necessary to travel, and to travel one must have money, and in order to get money you have to work.… I am in a vicious circle … from which it is impossible to escape.
>
> —Honoré Balzac,[4] 19th century novelist and playwright

> Here we meet a very important feature. It would seem as if this were circular reasoning; profits fell because investment fell, and investment fell because profits fell.
>
> —Jan Tinbergen,[5] economist

The second kind of feedback loop is amplifying, reinforcing, self-multiplying, snowballing—a vicious or virtuous circle that can cause healthy growth or runaway destruction. It is called a **reinforcing feedback loop**, and will be noted with an R in the diagrams. It generates more input to a stock the more that is already there (and less input the less that is already there). A reinforcing feedback loop enhances whatever direction of change is imposed on it.

For example:

- When we were kids, the more my brother pushed me, the more I pushed him back, so the more he pushed me back, so the more I pushed him back.
- The more prices go up, the more wages have to go up if people are to maintain their standards of living. The more wages go up, the more prices have to go up to maintain profits. This means that wages have to go up again, so prices go up again.
- The more rabbits there are, the more rabbit parents there are to make baby rabbits. The more baby rabbits there are, the more grow up to become rabbit parents, to have even more baby rabbits.
- The more soil is eroded from the land, the less plants are able to grow, so the fewer roots there are to hold the soil, so the more soil is eroded, so less plants can grow.
- The more I practice piano, the more pleasure I get from the sound, and so the more I play the piano, which gives me more practice.

Reinforcing loops are found wherever a system element has the ability to reproduce itself or to grow as a constant fraction of itself. Those elements include populations and economies. Remember the example of the interest-bearing bank account? The more money you have in the bank, the more interest you earn, which is added to the money already in the bank, where it earns even more interest.

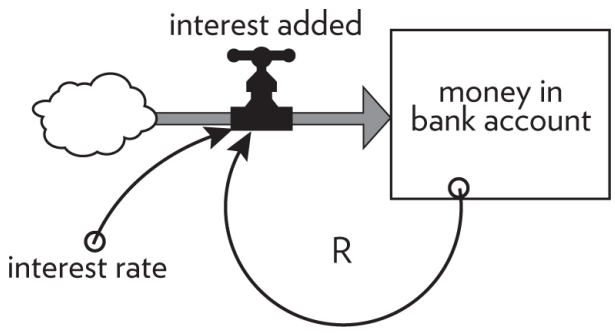


Figure 12. Interest-bearing bank account.

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Figure 13 shows how this reinforcing loop multiplies money, starting with $100 in the bank, and assuming no deposits and no withdrawals over a period of twelve years. The five lines show five different interest rates, from 2 percent to 10 percent per year.

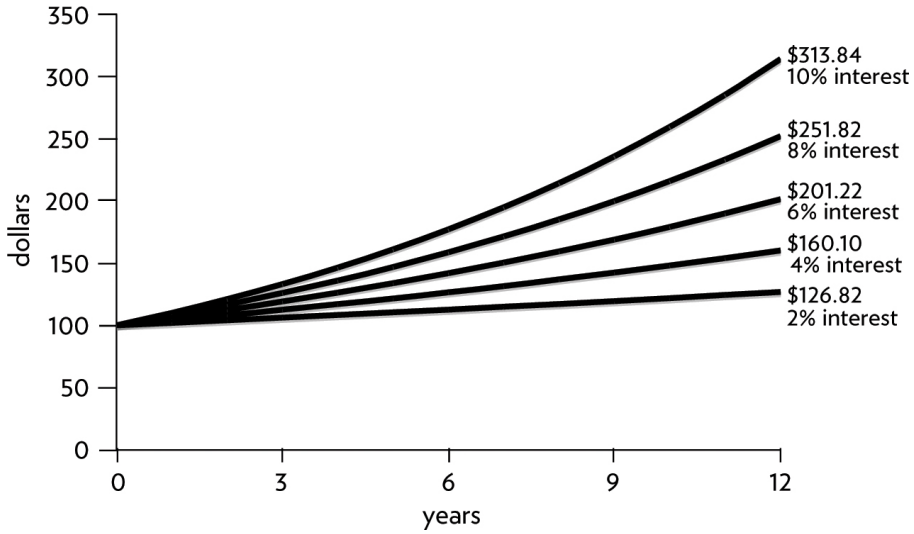


Figure 13. Growth in savings with various interest rates.

This is not simple linear growth. It is not constant over time. The growth of the bank account at lower interest rates may look linear in the first few years. But, in fact, growth goes faster and faster. The more is there, the more is added. This kind of growth is called "exponential." It's either good news or bad news, depending on what is growing—money in the bank, people with HIV/AIDS, pests in a cornfield, a national economy, or weapons in an arms race.

Reinforcing feedback loops are self-enhancing, leading to exponential growth or to runaway collapses over time. They are found whenever a stock has the capacity to reinforce or reproduce itself.

In Figure 14, the more machines and factories (collectively called "capital") you have, the more goods and services ("output") you can produce. The more output you can produce, the more you can invest in new machines and factories. The more you make, the more capacity you have to make even more. This reinforcing feedback loop is the central engine of growth in an economy.

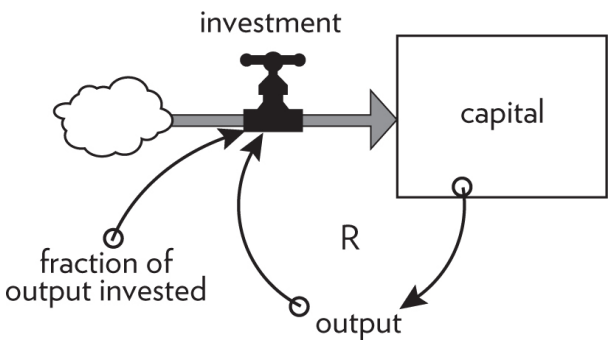


Figure 14. Reinvestment in capital.

By now you may be seeing how basic balancing and reinforcing feedback loops are to systems. Sometimes I challenge my students to try to think of any human decision that occurs *without* a feedback

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loop—that is, a decision that is made without regard to any information about the level of the stock it influences. Try thinking about that yourself. The more you do, the more you'll begin to see feedback loops everywhere.

The most common "non-feedback" decisions students suggest are falling in love and committing suicide. I'll leave it to you to decide whether you think these are *actually* decisions made with no feedback involved.

HINT ON REINFORCING LOOPS AND DOUBLING TIME

Because we bump into reinforcing loops so often, it is handy to know this shortcut: The time it takes for an exponentially growing stock to double in size, the "doubling time," equals approximately 70 divided by the growth rate (expressed as a percentage).

Example: If you put $100 in the bank at 7% interest per year, you will double your money in 10 years (70 ÷ 7 = 10). If you get only 5% interest, your money will take 14 years to double.

Watch out! If you see feedback loops everywhere, you're already in danger of becoming a systems thinker! Instead of seeing only how A causes B, you'll begin to wonder how B may *also* influence A—and how A might reinforce or reverse itself. When you hear in the nightly news that the Federal Reserve Bank has done something to control the economy, you'll also see that the economy must have done something to affect the Federal Reserve Bank. When someone tells you that population growth causes poverty, you'll ask yourself how poverty may cause population growth.

THINK ABOUT THIS:

If A causes B, is it possible that B also causes A?

You'll be thinking not in terms of a static world, but a dynamic one. You'll stop looking for who's to blame; instead you'll start asking, "What's the system?" The concept of feedback opens up the idea that a system can cause its own behavior.

So far, I have limited this discussion to one kind of feedback loop at a time. Of course, in real systems feedback loops rarely come singly. They are linked together, often in fantastically complex patterns. A single stock is likely to have several reinforcing and balancing loops of differing strengths pulling it in several directions. A single flow may be adjusted by the contents of three or five or twenty stocks. It may fill one stock while it drains another and feeds into decisions that alter yet another. The many feedback loops in a system tug against each other, trying to make stocks grow, die off, or come into balance with each other. As a result, complex systems do much more than stay steady or explode exponentially or approach goals smoothly—as we shall see.

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* Definitions of words in bold face can be found in the Glossary.

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